

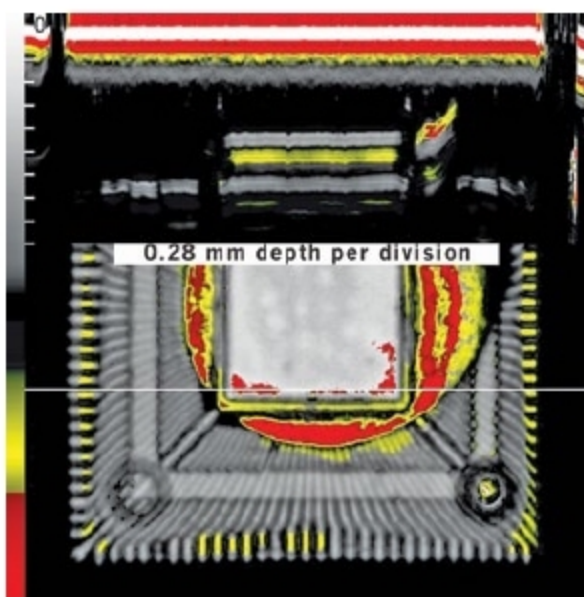


Fig. 3: Acoustic section through the line shows die face delamination and the crack profile

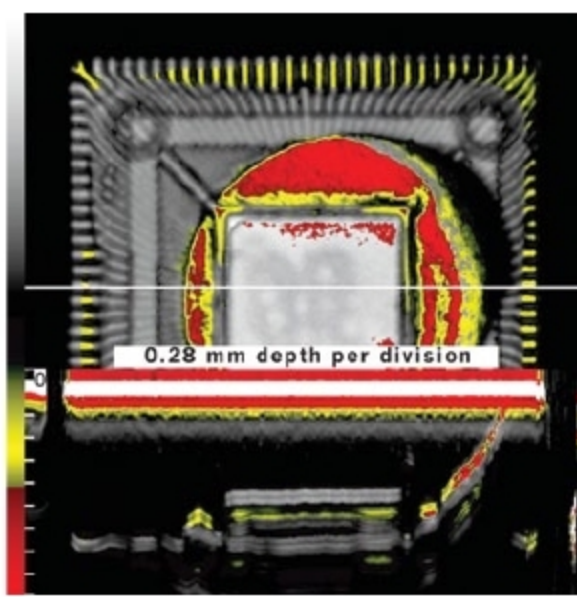


Fig. 4: Along the line, the crack rises above the gate and reaches the package's surface

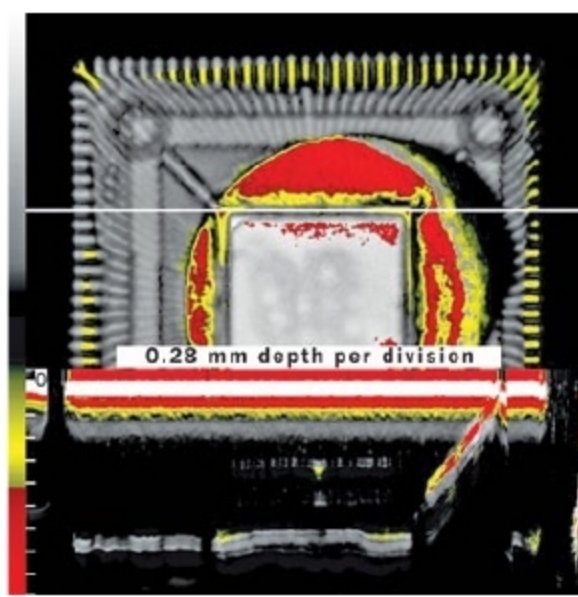


Fig. 5: Much of the crack is above the gate here

back-and-forth scans and altitude adjustments, the transducer reaches the top of the sample and the cross-sectional image is complete. The image will be dimensionally identical to an optical image made after sawing through the sample along the line and then photographing the exposed face. The same internal features, including defects, will be in the same places.

Fig. 1 is the top-view image of a plastic-encapsulated IC that has suffered a popcorn crack. Popcorn cracks typically originate in or near the die attach material and expand downwards or (as here) upwards. This image was gated on a depth from just above the die face to just below the die attach. Note that the gate from which echoes are accepted does not extend all the way to the top surface of the package.

The circular red-and-yellow feature is the top surface of the upwardly-slanted popcorn crack. Red-and-yellow features are gaps. On the face of the die, small red areas are horizontal delaminations between the die and the mould compound.

In Fig. 2, the Q-BAM cross-sectional image through the package is at the top and replaces part of the C-mode image. For vertical measurements, the 10 bars to the left of the Q-BAM image divide the thickness of the scan depth into 10 segments, each 0.028mm in vertical extent. The horizontal white line near the bottom of the C-mode image marks the vertical plane from the bottom to the top of which the

transducer collected return echo signals.

Starting at the left side of the C-mode image, the white line passes through the lead finger, tape and tie bar. These features reflect little ultrasound and can be seen as a very faint sequence of features in the Q-BAM image at the top. Farther along, the white line passes through the red popcorn crack; the Q-BAM image of the crack shows that, along this line, the right end of the crack extends upwards. Beyond this point only a few faint structures are visible.

The white line passes through the die face in Fig. 3. The small red features in the C-mode image are small (but dangerous) delaminations between the die face and the mould compound. These appear yellow in the Q-BAM image. To the right of the die, the popcorn crack extends a short distance upwards.

The Q-BAM image has been moved to the bottom in Fig. 4 to avoid impinging on the white line. The lead finger and tie bar reflections are faintly visible here, as is a part of the popcorn crack at the left of the die. At the right in the Q-BAM image, the popcorn crack can be seen extending all the way to the top of the package. In the C-mode image at the top, the crack does not appear to extend so far to the right, but a close look at the C-mode image shows a dark region to the right of the yellow crack. The crack extends above the top of the gated depth, but the black region is that vertical portion of the

gate from which no echoes were received.

In Fig. 5, the white line runs along the very edge of the die. The Q-BAM image reveals—faintly—the die face delaminations, as well as the upward crack at the right. The crack here reaches the surface above and once again is in part displayed in the C-mode image as a black area from which no echoes were received. Note that the two mould marks at the top corners of the package are also black because these lie above the gate.

The IC package imaged here by the Q-BAM method could be imaged along other lines parallel to these lines, if desired, or the package could be rotated and imaged along a line in any horizontal direction. Two advantages of the method are lack of damage to the part and sensitivity of ultrasound to very thin gaps. In C-mode imaging, a delamination for only a small fraction of a micron thick shows up strongly. In Q-BAM, the cross-section through the same delamination will show up just as strongly.

Q-BAM is sometimes used in conjunction with destructive physical analysis, often because a very thin gap may be difficult or impossible to see optically after physical sectioning. It is frequently used to plan locations for destructive physical analysis. C-mode and Q-BAM techniques image and locate the gap in three dimensions before physical sectioning is performed. **EFY**